

Feature Engineering, Model Optimization & Performance Comparison

House Price Predictor — California Housing Dataset

Task 2 | AI/ML Internship — Maincrafts Technology | Dhanushya Somu

1. Objective

Building on Task 1, this task focused on how Machine Learning engineers improve models in real-world projects. The goal was to apply feature scaling, train multiple regression algorithms, and compare their performance objectively to select the best-performing model for predicting median house prices on the California Housing dataset.

2. Methodology

The same California Housing dataset (20,640 records, 8 features) from Task 1 was used. The workflow followed these steps:

- **Feature scaling:** Applied *StandardScaler* to bring all 8 input features to a common scale (mean 0), preventing features with larger numeric ranges (e.g. Population) from dominating the model unfairly.
- **Train/test split:** 80% training, 20% testing, using *random_state=42* for reproducibility — identical split strategy as Task 1.
- **Models trained:** Linear Regression (baseline), Ridge Regression (alpha=1.0, penalizes large coefficients to reduce overfitting), and a Decision Tree Regressor (max_depth=5, captures non-linear relationships).
- **Evaluation:** Each model was scored on the same test set using RMSE (Root Mean Squared Error) and R² Score.

3. Model Performance Comparison

Model	RMSE	R ² Score
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree (max_depth=5)	0.7242	0.5997

The **Decision Tree Regressor achieved the best performance**, with the lowest RMSE (0.7242, ~\$72,400 average error) and the highest R² Score (0.5997, explaining ~60% of price variation). Linear Regression and Ridge Regression performed almost identically, since Ridge's penalty term had minimal effect at alpha=1.0 on this dataset — indicating the linear model was not significantly overfitting to begin with.

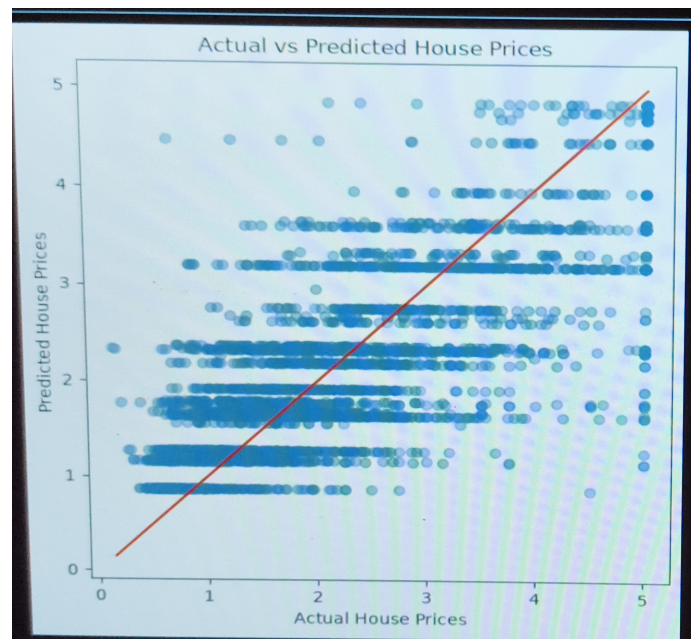


Figure 1: Actual vs. predicted house prices using the selected Decision Tree Regressor. The red line represents perfect prediction. Predictions cluster into horizontal bands, a characteristic pattern of Decision Tree models, which output a fixed value for each leaf node rather than a continuous range. Notably, the extreme outlier seen with Linear Regression in Task 1 does not appear here, since the Decision Tree cannot predict values outside the range seen during training.

4. Selected Model & Justification

The **Decision Tree Regressor** (max_depth=5) was selected as the final model. It outperformed both linear-based models on both evaluation metrics, indicating that some relationships between features (e.g. location, income) and house price are non-linear and are better captured by a tree-based split structure than by a straight-line equation. The max_depth was limited to 5 to keep the tree simple and reduce the risk of overfitting to the training data.

5. Key Learnings

- **Feature scaling** is essential when features exist on very different numeric ranges, ensuring fair learning across all inputs.
- **Comparing multiple models** objectively, rather than relying on a single algorithm, is standard practice in real ML workflows.
- **Ridge Regression's** regularization had negligible impact here, showing that regularization benefits depend on how much a model is initially overfitting.
- **Decision Trees** can outperform linear models when relationships in the data are non-linear, though deeper trees would need to be checked for overfitting on unseen data.

6. Conclusion

Three regression models were trained and evaluated on the scaled California Housing dataset. The Decision Tree Regressor was selected as the best-performing model, improving R^2 from 0.576 (Task 1's baseline Linear Regression) to 0.600. This task demonstrated the value of feature scaling, systematic model comparison, and metric-driven decision-making in a real machine learning workflow.